AI / ML ENGINEER

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SUMMARY

Final-year AI student at Multimedia University focused on engineering automated data pipelines and applying ML/DL to real-world datasets. Engineered and shipped an end-to-end ETL system processing 1,470 records with zero manual intervention; experienced in optimized SQL (CTEs, window functions) and production-style ML workflows. Seeking an AI or ML engineering internship.

TECHNICAL SKILLS

Languages & ML/AI: Python, SQL, C++, Machine Learning, Deep Learning, LSTM, XGBoost, Random Forest, Feature Engineering, Model Evaluation

Data Engineering & Tools: ETL Pipeline Design, SQLite, Pandas, NumPy, TensorFlow/Keras, Scikit-learn, Streamlit, Matplotlib, Power BI, Google Analytics

PROJECTS

AI-Powered Autonomous Supply Chain Optimization & Risk Management System (SCOMS) — In Progress, Expected Mar 2027

Final Year Project, B.CS (AI), MMU · Python, TensorFlow/Keras, Scikit-learn, XGBoost, Streamlit

- Designed a three-module AI pipeline integrating LSTM demand forecasting, Random Forest supplier risk classification, and EOQ/ROP inventory optimization for retail supply chain decision support
- Architected LSTM forecasting model (2 stacked layers, 64/32 units, 30-day sliding window) for 7–30 day demand prediction; configured XGBoost benchmark across RMSE, MAE, and MAPE evaluation metrics
- Built Random Forest classifier (100 estimators) scoring suppliers as Low/Medium/High risk across five operational features: delivery delay rate, fulfillment rate, lead-time variance, defect rate, disruption history
- Implemented inventory module computing EOQ, Reorder Point, and Safety Stock dynamically from forecast output; surfaced via Streamlit dashboard with role-based recommendation flow for store managers
- Finalized system architecture and validated dataset strategy across 5-chapter FYP research phase, selecting a hybrid LSTM + XGBoost approach — benchmarked against 11 peer-reviewed supply chain optimization papers (2020–2026) to justify model selection over RNN and ARIMA baselines

Automated HR Data Pipeline & Predictive Risk Engine — Feb 2026

Python, SQL, SQLite, Pandas, openpyxl

- Architected a fully automated ETL pipeline ingesting 1,470 employee records — Python handled extraction, schema validation, and SQLite loading with zero manual data entry
- Engineered 3 composite flight-risk features from 14 raw HR attributes — including tenure-to-salary ratio and overtime frequency — enabling downstream SQL queries to surface 113 high-risk employees (7.7% of the 1,470-record dataset) with structured, reproducible logic
- Optimized SQL transformation layer using CTEs, window functions, and partitioned aggregations to rank, filter, and join across normalized employee tables
- Surfaced retention signal: Sales Representatives exhibited 39.76% attrition at lowest average salary ($2,626) — validated through SQL aggregation
- Published full pipeline codebase on GitHub with structured README and reproducible execution steps

XGBoost Diabetes Risk Models — Clinical & Behavioral Datasets — 2025

Python, XGBoost, SMOTE, GridSearchCV, Scikit-learn

- Built two parallel binary classification pipelines predicting diabetes onset — 2,000-patient clinical dataset (Glucose, BMI, BloodPressure, Insulin, Age) and a 50/50 balanced 13-feature behavioral/socioeconomic dataset
- Engineered 2 clinical interaction features (BMI×Age, Glucose/Insulin ratio); applied median imputation across 5 columns (956 Insulin, 573 SkinThickness nulls) before training
- Resolved class imbalance through SMOTE oversampling applied exclusively to the training split, lifting minority-class F1 to 0.76 — ensuring the model remained clinically viable for detecting at-risk patients rather than defaulting to majority-class prediction
- Executed 5-fold GridSearchCV across 432 hyperparameter combinations — optimal config (max_depth=7, n_estimators=300, lr=0.05) achieved 82% test accuracy on minority class
- Behavioral pipeline used stratified splits and StandardScaler normalization, achieving 79.4% accuracy with balanced precision/recall (0.79/0.80) — validated via confusion matrices on all splits

EXPERIENCE

Committee Member, Activities Division — Mar 2025 – Mar 2026

Arabic Culture Society — Multimedia University (Part-time)

- Coordinated end-to-end planning and on-site execution of multiple student engagement events across a 12-month term, working within a cross-functional committee structure
- Managed event logistics — scheduling, vendor and venue coordination, internal communications, and post-event reviews — and contributed to outreach strategy to grow attendance

EDUCATION

Multimedia University, Melaka, Malaysia — Mar 2024 – Mar 2027

Bachelor of Computer Science (Artificial Intelligence) · GPA: 3.4 · Dean's List

Relevant Coursework: Machine Learning, Deep Learning, AI Fundamentals, Data Analytics, Database Systems, Data Structures & Algorithms, Computer Programming (C++), Software Engineering

CERTIFICATIONS & ADDITIONAL INFORMATION

HCIA-AI V4.0 — Huawei | CCNA: Introduction to Networks | Claude Code 101 — Anthropic | Microsoft Power BI for Beginners | Google Analytics Certification

Languages: Arabic (Native), English (Professional working proficiency)